Task C — Clustering and PCA

1. Objective

The objective of Task C is to identify meaningful booking segments in the City Hotel (H2) dataset using unsupervised machine learning. K-Means was selected as the initial baseline clustering algorithm, while PCA was used to visualize the resulting clusters.

2. Dataset Understanding and Cleaning

The original dataset contains 79,330 records and 31 variables. The following preprocessing steps were performed:

- Inspected data types, missing values, duplicates and summary statistics.
- Cleaned categorical text values and handled missing-value markers.
- Removed 25,902 exact duplicate records.
- Removed bookings with zero guests. The original dataset contained 167 such records, of which 154 remained after duplicate removal.
- Created **TotalNights = StaysInWeekendNights + StaysInWeekNights**.
- Removed 231 zero-night bookings from the remaining records.
- Checked selected numerical features for negative values.
- Investigated distributions, skewness and outliers.

After cleaning, **53,043 records** remained for clustering.

3. Exploratory Data Analysis

EDA was performed using histograms, boxplots, a correlation heatmap and descriptive statistics.

The histograms and boxplots were used to examine numerical feature distributions and identify extreme observations. Skewness was examined for LeadTime, TotalNights, Adults, Children and IsRepeatedGuest.

The correlation heatmap was used to investigate redundancy among numerical clustering features. Category cardinality and frequency distributions were also examined.

Extreme values in LeadTime and TotalNights were capped at their respective 99th percentiles within the preprocessing pipeline.

4. Feature Selection

Eight features were selected for clustering:

Feature	Justification
LeadTime	Represents how early a booking was made
TotalNights	Represents the length of the stay
Adults	Describes the number of adult guests
Children	Helps distinguish family bookings
Meal	Describes the selected meal package
MarketSegment	Represents the booking market segment
IsRepeatedGuest	Distinguishes returning and first-time guests
ReservedRoomType	Represents the room type originally reserved

Feature selection was based on booking relevance and an unsupervised correlation filter with a threshold of 0.90. A zero-variance filter was also applied after encoding.

Final reservation outcomes and post-booking variables were excluded to avoid leakage and maintain the unsupervised nature of clustering.

5. Preprocessing Pipeline

A reusable scikit-learn pipeline was created with the following steps:

1. Cap extreme LeadTime and TotalNights values at the 99th percentile.
2. Replace missing numerical values using the median.
3. Replace missing categorical values using the most frequent category.
4. Scale numerical variables using StandardScaler.
5. Convert categorical variables using OneHotEncoder.
6. Remove constant encoded features using VarianceThreshold.

The pipeline produced 24 processed features and no remaining missing values.

6. K-Means Baseline Model

K-Means was selected as the baseline clustering algorithm.

The Elbow Method was used to investigate cluster counts from K = 2 to K = 10. Silhouette Score and Davies–Bouldin Index were also calculated for K values from 2 to 6.

The initial baseline was trained using K = 3, a fixed random seed of 42 and n_init = 10.

Clustering Results

K	Silhouette Score	Davies–Bouldin Index
2	0.563	0.567
3	0.201	1.609
4	0.219	1.387
5	0.239	1.313
6	0.249	1.209

The K = 3 baseline achieved a Silhouette Score of 0.201 and a Davies–Bouldin Index of 1.609.

K = 2 achieved better values on both metrics. However, K = 3 was retained as the initial baseline experiment for Milestone 1. Final cluster-count selection will be revisited during model optimization.

The Silhouette Score was calculated on a reproducible random sample of 2,000 observations for computational efficiency.

7. PCA Visualization

PCA was applied to the preprocessed features to visualize the K-Means clusters in two dimensions.

The first principal component explained approximately 24.2% of the variance, while the second explained 16.4%, giving approximately 40.5% combined explained variance.

A two-dimensional scatter plot was produced using the PCA coordinates and the K-Means cluster labels.

PCA was used for visualization only; K-Means was trained on the full preprocessed feature representation.

8. Cluster Interpretation

The K = 3 model identified three booking groups:

Cluster	Bookings	Average LeadTime	Average TotalNights
0	21,844	140.03 days	4.33 nights
1	29,657	35.80 days	2.36 nights
2	1,542	12.33 days	1.81 nights

Cluster 0 — Early bookings and longer stays: This cluster is characterized by relatively long booking lead times and longer stays.

Cluster 1 — Shorter stays and later bookings: This cluster contains bookings with shorter stays and considerably shorter lead times than Cluster 0.

Cluster 2 — Repeated guests: This cluster contains the repeated-guest records, with shorter average lead times and stays. The IsRepeatedGuest feature has a strong influence on the segmentation.

The clusters show differences in booking behavior, although the relatively low Silhouette Score for K = 3 indicates limited overall separation.

9. Preliminary Leakage Audit

Clustering was performed without using IsCanceled or ReservationStatus as input variables.

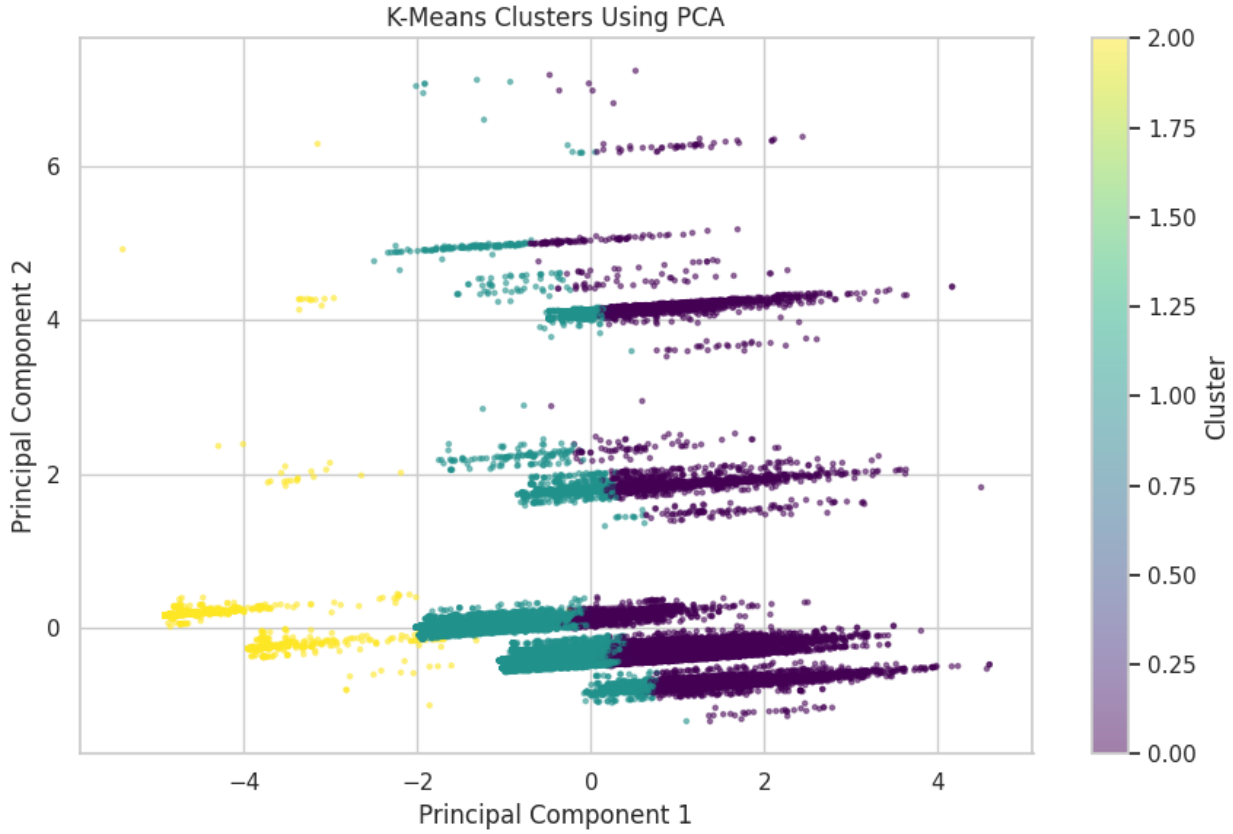
Variables representing final outcomes or post-booking information, including ReservationStatusDate and AssignedRoomType, were excluded.

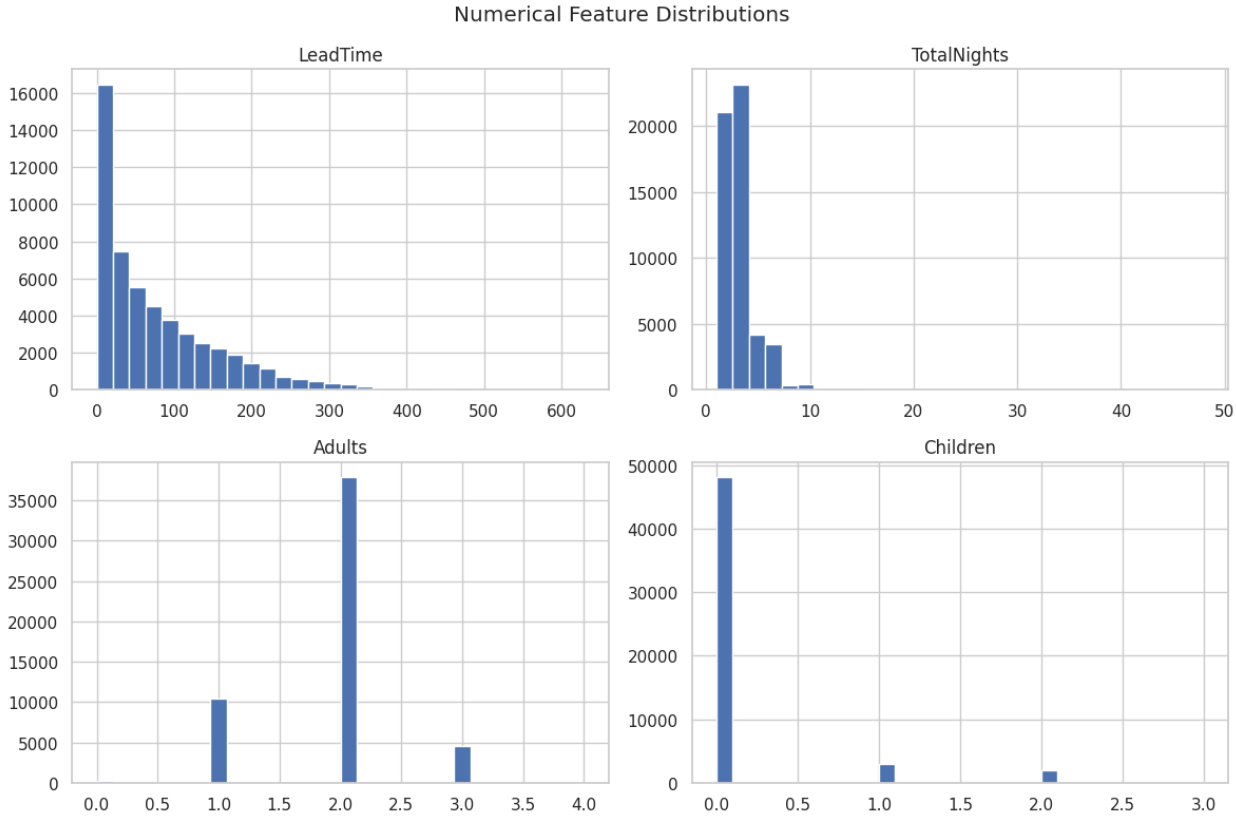
The selected variables describe booking characteristics rather than final booking results. Although IsRepeatedGuest is not a final-outcome variable, its strong influence on cluster formation was identified as a limitation to investigate later.

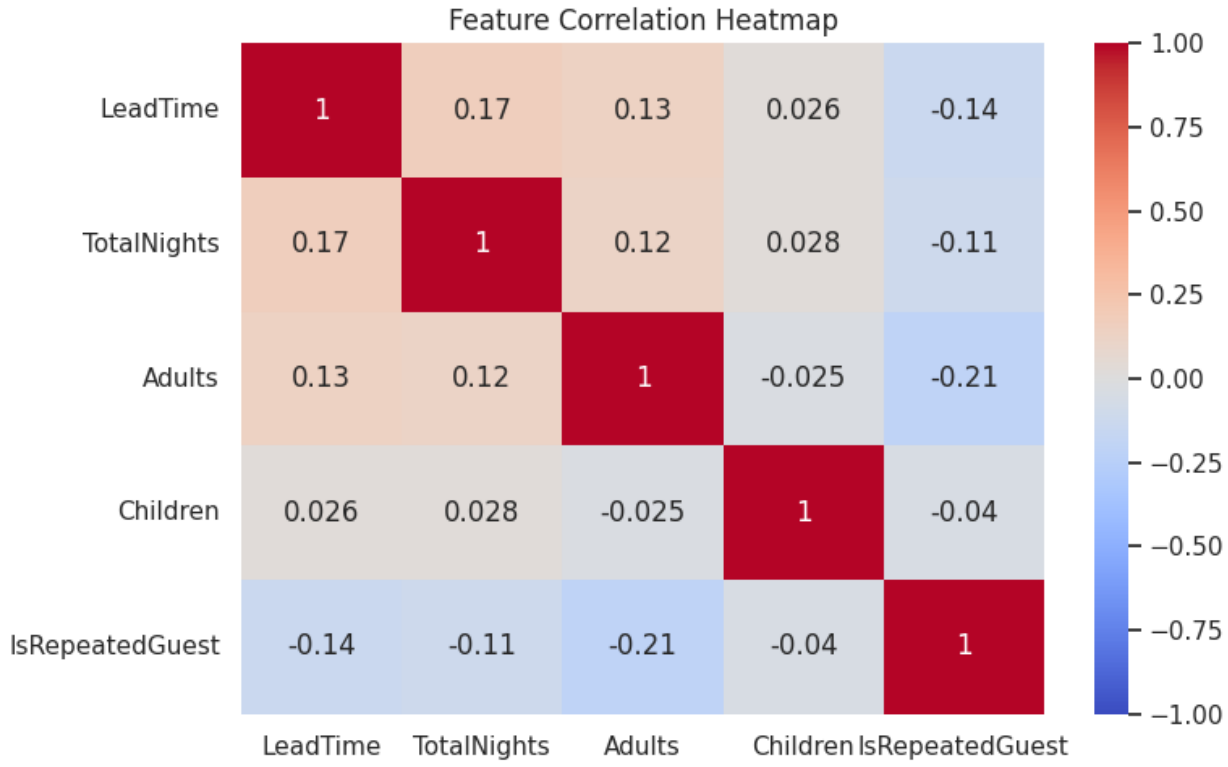
10. Phase 2 Plan

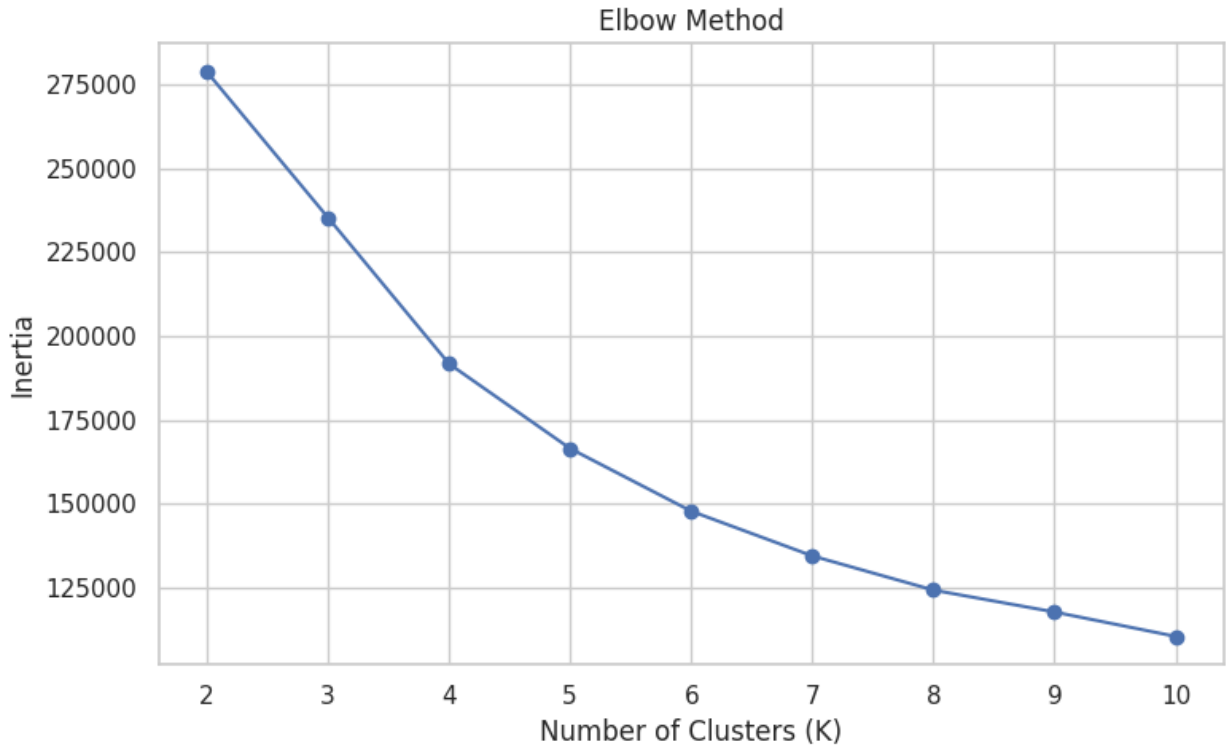
The next milestone will include:

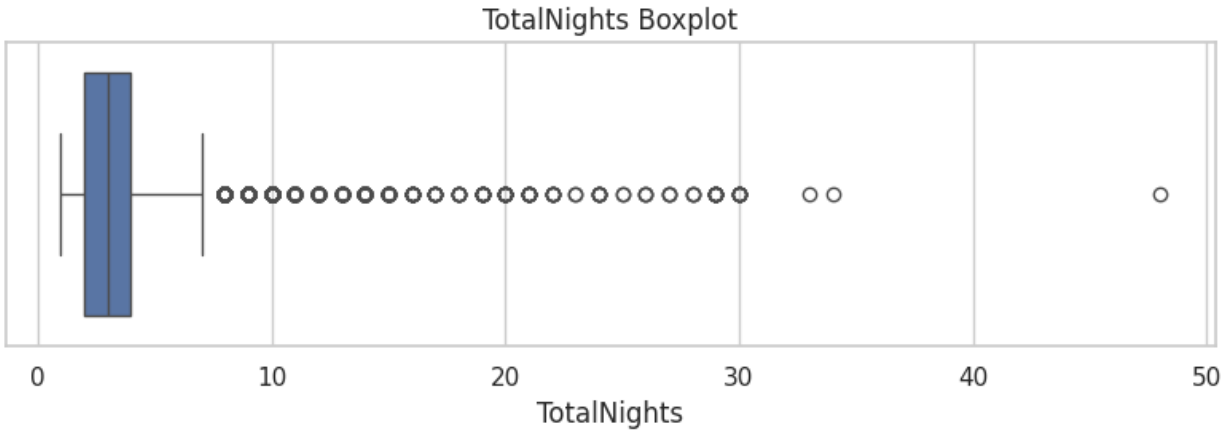
- Comparing K-Means with additional clustering algorithms.
- Performing systematic parameter optimization.
- Revisiting the cluster count and feature selection.
- Completing the required PCA/LDA experiments in coordination with the classification workstream.
- Selecting and integrating the final clustering solution into the end-to-end application.

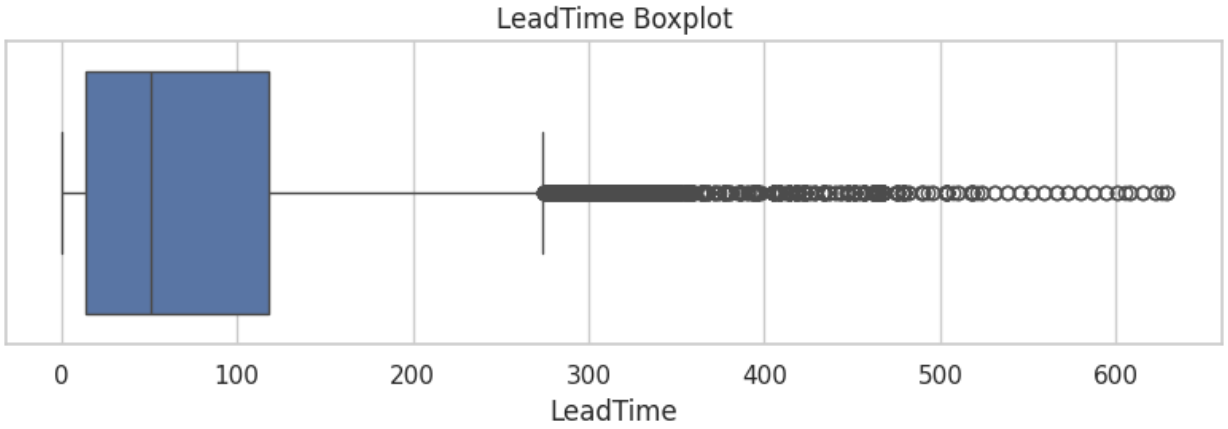

K-Means Clusters Using PCA
Principal Component 2
Principal Component 1
Cluster
6
4
2
0
−4
−2
0
2
4
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00

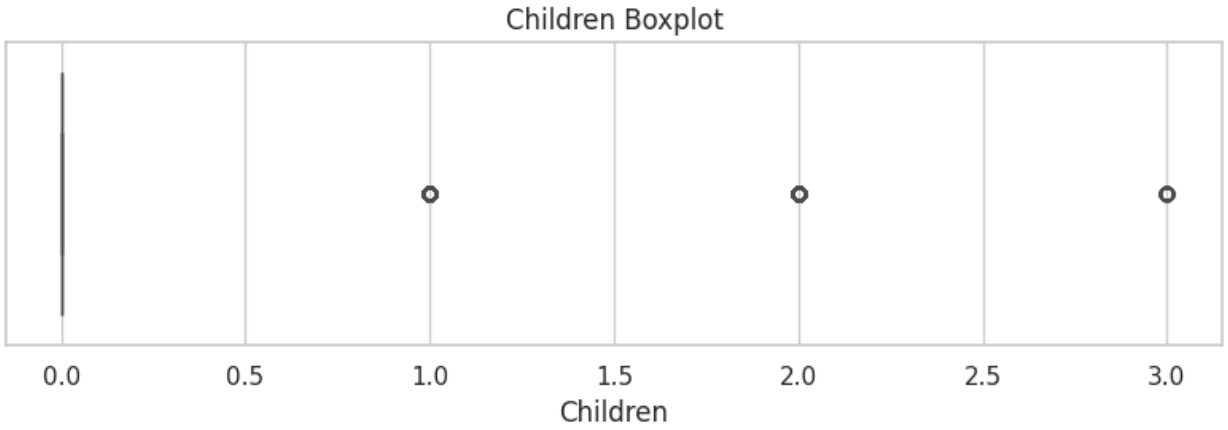

Numerical Feature Distributions
LeadTime
TotalNights
Adults
Children

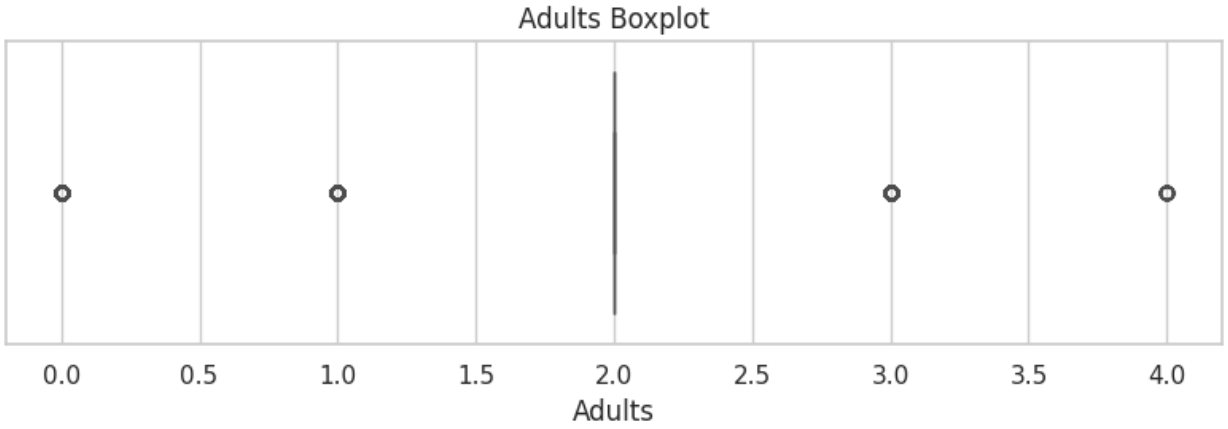

Feature Correlation Heatmap
LeadTime
TotalNights
Adults
Children
IsRepeatedGuest
1
0.17
0.13
0.026
-0.14
0.17
1
0.12
0.028
-0.11
0.13
0.12
1
-0.025
-0.21
0.026
0.028
-0.025
1
-0.04
-0.14
-0.11
-0.21
-0.04
1
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

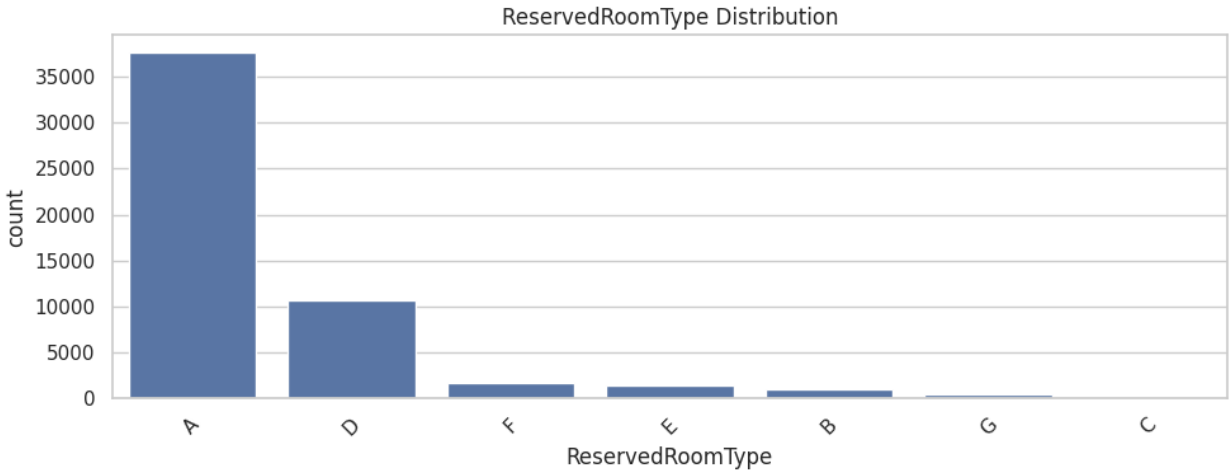

Elbow Method
Inertia
275000
250000
225000
200000
175000
150000
125000
2
3
4
5
6
7
8
9
10
Number of Clusters (K)

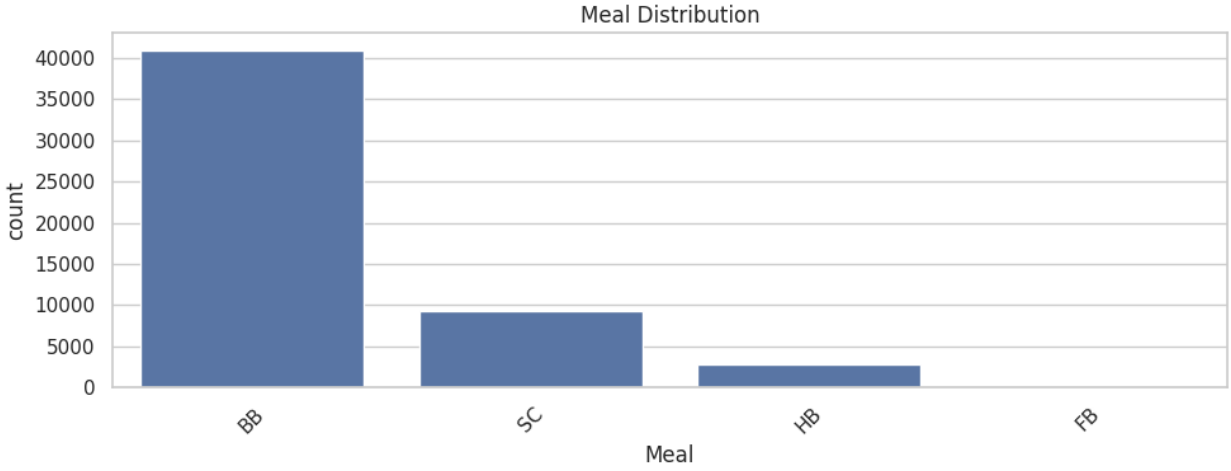

TotalNights Boxplot
0
10
20
30
40
50
TotalNights

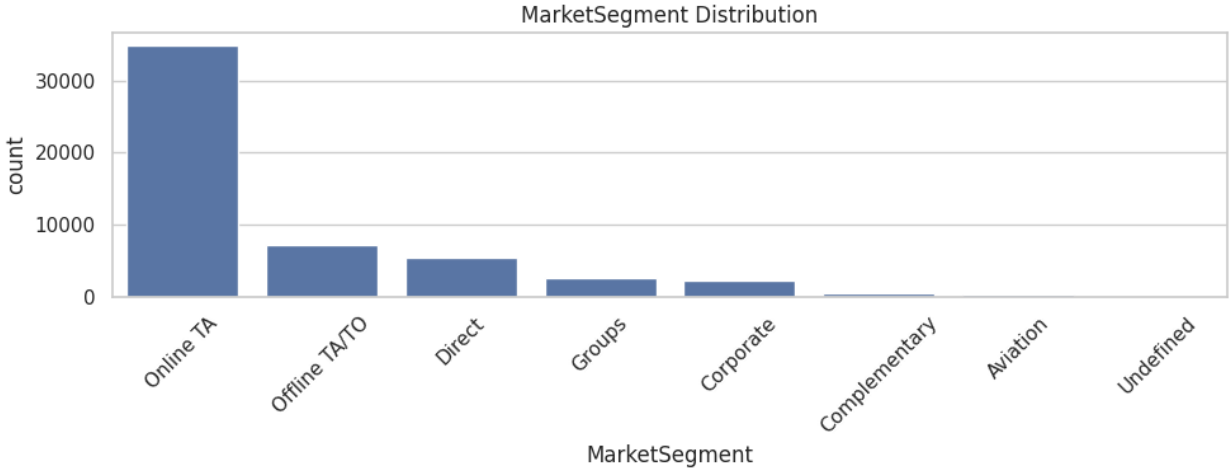

LeadTime Boxplot
0
100
200
300
400
500
600
LeadTime


Children Boxplot
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Children


Adults Boxplot
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Adults


ReservedRoomType Distribution
count
35000
30000
25000
20000
15000
10000
5000
0
A
D
F
E
B
G
C
ReservedRoomType


Meal Distribution
count
40000
35000
30000
25000
20000
15000
10000
5000
0
BB
SC
HB
FB
Meal


MarketSegment Distribution
count
30000
20000
10000
0
Online TA
Offline TA/TO
Direct
Groups
Corporate
Complementary
Aviation
Undefined
MarketSegment